

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, Go, Java, C++, JavaScript, TypeScript, SQL, Bash
Systems & Performance	Distributed data systems, Spark / PySpark, Spark UI, Linux profiling, async services / queues, integration & load testing, OpenMP / SIMD concepts
Backend & Reliability	Node.js, Express, FastAPI, REST APIs, authentication hardening, rate limiting, audit logging, multi-tenant SaaS, VAPT remediation
Cloud & DevOps	Docker, Kubernetes, AWS, GCP, Azure, CI/CD, Git, GitHub Actions, Cloudflare Workers (personal project)
Databases & Tools	PostgreSQL, MySQL, MongoDB, Redis, SQLite, GTest / CMake

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability for **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Debugged complex, tenant-specific issues across backend services and third-party integrations by monitoring production logs during releases; improved operational stability through authentication hardening, rate limiting, and audit logging.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical data against regressions, writing maintainable, secure-by-default code across tenant isolation and backend access paths.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated a core codebase from JavaScript to **TypeScript** across multiple repositories, improving type safety and maintainability of a large, evolving system.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a Spark ETL pipeline on the NYC TLC taxi dataset by tuning broadcast joins, caching, partitioning, and adaptive query-execution settings to cut shuffle volume and wall-clock runtime across distributed workers.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks, then applied column pruning and precomputed fields to reduce data movement, documenting a repeatable performance-optimization workflow.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and individual tool failures; conversation and task state persist server-side in D1 without an external database.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Hardened a multi-stage web-discovery service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation for observability and review-friendly debugging.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Engineered FastAPI backend and Electron/React frontend to run **fully offline**; designed async APIs and queues to keep the UI responsive while processing local video and audio streams on device.

Publication & Extracurriculars

- Open Source:** Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed features to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.